

Can Anything Beat VIX? A Systematic Out-of-Sample Evaluation of Eleven Signal Families for Equity Volatility Forecasting and Volatility Timing*

Yi-Hao Lai[†] VolPred Research System

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Abstract

We conduct a systematic out-of-sample horse race among eleven pre-specified signal families—cross-asset volatility momentum, VIX term structure, behavioral sentiment, the variance risk premium, multi-asset portfolio optimization, equal risk contribution, Bitcoin volatility, the yield curve slope, Google Trends fear proxies, overnight VIX changes, and calendar anomalies—evaluating whether any can improve upon VIX alone for equity volatility forecasting and volatility-timing portfolio construction. Using daily S&P 500 data spanning 1993–2026, we apply a unified forecast pipeline with strict lag enforcement, transaction-cost accounting, the [Harvey et al. \(2016\)](#) multiple-testing threshold ($|t| > 3.0$), formal Holm-Bonferroni correction, and proxy-robust evaluation following [Patton \(2011\)](#). The main finding is strongly negative: not a single signal family produces a statistically significant out-of-sample improvement—whether measured by the standard [Campbell and Thompson \(2008\)](#) R^2_{OOS} relative to the historical mean, by incremental R^2 relative to VIX-only forecasts, or by Sharpe ratio gains for volatility-timing strategies—and all results survive Holm-Bonferroni family-wise error rate correction. The VIX–realized

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[†]Department of Finance, Da-Yeh University. Corresponding author. Email: yhlai@mail.dyu.edu.tw

volatility R^2 ranges from 0.24 to 0.64 across five non-overlapping eras spanning 33 years (coefficient of variation = 0.37), demonstrating time-invariance. Volatility model rankings are criterion-dependent—GJR-GARCH dominates under proxy-robust QLIKE on r_t^2 (Model Confidence Set sole member) while AMEM dominates for VaR/ES risk management (composite score 1.94 vs. 1.63)—yet VIX sufficiency holds regardless of which model or criterion is used. We frame VIX sufficiency as a rational outcome of option-market information aggregation and show that volatility timing functions as drawdown insurance—costly in Sharpe terms (average drag 3.49%/year for 12/VIX) but welfare-improving for investors with CRRA risk aversion $\gamma \geq 4.5$. The aggregate null conceals structured channel-specific heterogeneity: HAR+VIX achieves regime-invariant gains on Parkinson realized volatility for large-cap equity (QQQ/IWM, all three VIX regimes), GARCH-MIDAS long-run factors are universally null (0 conditional passes across 54 cells), and GAS-t score-driven models are architecturally harmful on equity (DM $t = -3.27$) yet provide validated tail-risk estimation on commodity (ES passes 4/4 assets). A seven-asset cross-asset Publication-In-Time (PIT) panorama spanning equities, commodities, fixed income, and cryptocurrency (SPY, GLD, TLT, BTC-USD, QQQ, USO, EEM) delivers UNIVERSAL_NULL_7/7: of 28 specification cells, only one isolated outlier (TLT $\times$ financial-stress, DM $t = +3.74$) reaches the Harvey threshold, collapsing to $t = +2.00$ under a one-week extra lag and failing the QLIKE materiality gate, confirming native-IV sufficiency as the broadest cross-asset finding available at daily frequency. The informative null sharpens the research frontier: future gains require either higher-frequency realized measures or genuinely exogenous information not already embedded in option prices.

Keywords: VIX, volatility forecasting, volatility timing, out-of-sample evaluation, variance risk premium, multiple testing, option-implied volatility, drawdown insurance, proxy-robust loss, model confidence set, Value-at-Risk

JEL Classification: C53, G11, G12, G17

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1 Introduction

The Chicago Board Options Exchange Volatility Index (VIX) is the single most widely used measure of expected equity market volatility. Derived from the prices of S&P 500 index options, VIX represents the market’s consensus forecast of 30-day annualized volatility and has been labeled the “fear gauge” of financial markets (Whaley, 2000). A vast literature documents that VIX is a powerful, though biased, predictor of future realized volatility (Christensen and Prabhala, 1998; Jiang and Tian, 2005; Bekaert and Hoerova, 2014). Yet the practical question facing portfolio managers, risk officers, and retail investors remains open: *can any observable signal improve upon VIX for forecasting equity volatility and timing equity exposure?*

This question is not merely academic. Volatility-timing (VT) strategies, which scale equity exposure inversely to expected volatility, have attracted substantial interest since the seminal work of Fleming et al. (2001) and the volatility-managed portfolio framework of Moreira and Muir (2017). If a signal can forecast volatility more accurately than VIX, it can potentially improve the risk-return trade-off of these strategies. Conversely, if VIX already aggregates all publicly available information relevant to near-term volatility, then the search for a “better VIX” is futile, and research efforts should be redirected.

The existing literature offers fragments of evidence but no systematic verdict. Individual studies have examined specific signal categories—the variance risk premium (Bollerslev et al., 2009), cross-asset spillovers (Diebold and Yilmaz, 2012), macroeconomic determinants of time-varying volatility (Schwert, 1989; Engle and Rangel, 2008), and sentiment measures (Baker and Wurgler, 2006)—but each study uses its own sample period, evaluation metric, and benchmark specification, making cross-study comparison unreliable. Moreover, most studies do not enforce the multiple-testing corrections that Harvey et al. (2016) argue are essential when many predictors are examined.

We address this gap by conducting the first comprehensive, pre-registered horse race among eleven signal families, evaluated under a unified framework with six methodological safeguards:

- (i) **Pre-specification:** All eleven signal families and their construction rules are defined before examining out-of-sample results, eliminating data-snooping in signal selection.
- (ii) **Strict lag enforcement:** Every signal is lagged one trading day (`signal.shift(1)`) before being applied to returns, preventing look-ahead bias—a surprisingly common error that we document has inflated Sharpe ratios by up to 373% in prior work.¹
- (iii) **Transaction costs:** All strategies deduct 5 basis points per leg on each weight change, applied to both buy and sell sides.
- (iv) **Multiple-testing control:** We adopt the approximate conservative threshold of $|t| > 3.0$ motivated by [Harvey et al. \(2016\)](#) and implement formal Holm-Bonferroni sequential correction across all eleven Diebold-Mariano tests, controlling the family-wise error rate at $\alpha = 0.05$.
- (v) **Proxy-robust evaluation:** Following [Patton \(2011\)](#), we evaluate forecasts using QLIKE on r_t^2 (a proxy-robust loss function), supplemented by Spearman rank correlations and the Model Confidence Set of [Hansen et al. \(2011\)](#), ensuring that our rankings are not artifacts of the volatility proxy choice.
- (vi) **Cross-era stability:** Results are evaluated separately across five non-overlapping eras spanning 1993–2026, preventing any single regime from driving conclusions.

Our main finding is strongly negative but highly informative. Across 11 signal families, 36 VIX sufficiency tests, and five market eras totaling 8,325 trading days, we find:

- No signal family produces a statistically significant out-of-sample R^2 improvement over VIX alone (maximum incremental $R^2 = 0.038$, all Diebold-Mariano $|t| < 3.0$).
- No signal-based strategy statistically outperforms the simple 12/VIX benchmark or the static 50/50 SPY/GLD allocation (maximum Sharpe improvement = +0.010, not significant).

¹In our own research program, Codex code review identified look-ahead bias in four separate experiments (K618, K621, K679, K698), with one case producing a Sharpe ratio of 1.68 that collapsed to 0.355 after lag correction.

- VIX’s forecasting power is time-invariant: the VIX–realized volatility R^2 ranges from 0.24 (low-volatility QE era, 2012–2020) to 0.64 (post-dot-com era, 2000–2007), with a coefficient of variation of only 0.37.
- Even the most economically motivated signal—the variance risk premium—achieves an in-sample t -statistic of 3.51 for return prediction but delivers out-of-sample $R^2 < 0.3\%$.
- The aggregate null conceals structured channel-specific heterogeneity: HAR+VIX achieves regime-invariant Parkinson-volatility gains on large-cap equity (QQQ/IWM, all three VIX regimes); GARCH-MIDAS long-run factors are universally null (0 of 54 conditional passes); GAS- t is architecturally harmful on equity (DM $t = -3.27$ for SPY) but provides validated tail-risk estimation on negatively-skewed commodities (ES backtest passes 4/4 assets at 1% level).
- Native-IV sufficiency extends universally across seven asset classes: of 28 cross-asset PIT specification cells (SPY, GLD, TLT, BTC-USD, QQQ, USO, EEM $\times$ four alt-data specs), only one cell (TLT $\times$ financial-stress, DM $t = +3.74$) crosses the Harvey threshold under the primary specification, and it fails every robustness check (lag-sensitive: collapses to $t = +2.00$ under one-week extra lag; QLIKE improvement = +0.50%, below the 5% materiality gate).

We interpret this null through the lens of option-market information aggregation. VIX is not simply a backward-looking statistical estimate; it is the forward-looking consensus of thousands of option traders who collectively incorporate cross-asset signals, macroeconomic data, sentiment, and term structure information into their pricing. The eleven signal families we test are precisely the inputs that option traders use. VIX, by construction, already contains them.

This interpretation leads to our second contribution: a reconceptualization of VT strategies. Rather than viewing VT as an alpha-generating timing strategy, we propose the *drawdown insurance* framework. VT sacrifices average returns (3.49%/year for 12/VIX, 2.12%/year for EWMA) in exchange for maximum drawdown reduction. This

trade-off is welfare-improving for investors with CRRA risk aversion $\gamma \geq 4.5$, making VT rational for moderately risk-averse investors even though it underperforms buy-and-hold on standard metrics.

Our third contribution concerns the research frontier. The informative null delineates what *cannot* work (daily signals already in the option-market information set) from what *might* work: genuinely exogenous shocks (e.g., geopolitical events, pandemic declarations) that arrive between option-market pricing updates, and possibly higher-frequency realized measures (5-minute data) whose relative performance is as yet unresolved. A small-sample pilot (K745, $N = 37$) directly comparing daily HAR-ABS (QLIKE = 0.077) with 5-minute HAR-RV (QLIKE = 0.109) found that the daily model *outperformed* the 5-minute model by 41.8% on this limited sample—the reverse of what the high-frequency literature conventionally predicts. This preliminary result is not powered to establish that 5-minute data cannot improve forecasts; it shows only that under the K745 specification and sample, 5-minute HAR-RV did not beat daily HAR-ABS. A larger-scale experiment (K1139, planned) will use an extended 5-minute SPY dataset to resolve whether 5-minute information carries incremental predictive content beyond daily closing-price VIX. We therefore flag the 5-minute frontier as open but not yet empirically established.

The remainder of the paper is organized as follows. Section 2 reviews the theoretical and empirical foundations of VIX as a benchmark. Section 3 describes the data and the eleven signal families. Section 4 details the forecast design, loss function choice, and the Patton (2011) proxy-robustness framework. Section 5 explains the statistical evaluation framework, including the Model Confidence Set of Hansen et al. (2011). Section 6 describes the volatility-timing strategy design. Section 7 presents the main results, criterion-dependent model rankings, and robustness checks. Section 8 discusses why the null is informative, including economic significance via VaR/ES backtesting. Section 9 concludes.

2 Why VIX is the Benchmark

2.1 Theoretical Foundations

The VIX index, introduced by the CBOE in 1993 and reformulated in 2003, measures the risk-neutral expected variance of the S&P 500 over the next 30 calendar days. Under the model-free framework of [Britten-Jones and Neuberger \(2000\)](#) and [Jiang and Tian \(2005\)](#), VIX squared equals the risk-neutral expectation of integrated variance:

$$\text{VIX}^2 = \frac{2}{\tau} \sum_i \frac{\Delta K_i}{K_i^2} e^{r\tau} Q(K_i) - \frac{1}{\tau} \left(\frac{F}{K_0} - 1 \right)^2, \quad (1)$$

where τ is time to expiration, K_i are strike prices, $Q(K_i)$ are midpoint option prices, F is the forward index level, and K_0 is the first strike below F . This construction aggregates information from the entire volatility smile, making VIX a sufficient statistic for the market's assessment of near-term volatility under risk-neutral pricing.

The link between VIX and future realized volatility (RV) is governed by the variance risk premium (VRP):

$$\mathbb{E}^Q[\sigma_{t,t+\tau}^2] = \mathbb{E}^P[\sigma_{t,t+\tau}^2] + \text{VRP}_t, \quad (2)$$

where $\mathbb{E}^Q$ and $\mathbb{E}^P$ denote risk-neutral and physical expectations, respectively. VIX overestimates realized volatility on average (the VRP is positive approximately 85% of days in our sample), but this bias is systematic and does not impair VIX's role as a forecasting benchmark—it merely shifts the intercept of the VIX–RV regression.

2.2 Empirical Evidence for VIX Forecasting Power

The volatility forecasting literature is vast; [Poon and Granger \(2003\)](#) provide the definitive survey, cataloguing over 90 studies and concluding that implied volatility generally dominates time-series models. Within this literature, [Christensen and Prabhala \(1998\)](#) show that VIX subsumes the information in historical volatility for forecasting realized volatility. [Jiang and Tian \(2005\)](#) demonstrate that model-free implied volatility dominates both GARCH and Black-Scholes implied volatility in encompassing regressions.

Bekaert and Hoerova (2014) decompose VIX into a conditional variance component and a variance risk premium, finding that the conditional variance component tracks realized volatility closely.

The realized volatility framework underpinning our forecast target originates with Andersen et al. (2003), who show that modeling and forecasting realized volatility—constructed from high-frequency returns—provides a natural benchmark for evaluating volatility forecasts. Our use of 22-day realized volatility as the forecast target follows this tradition. Patton (2011) demonstrates that loss function choice matters when the volatility proxy is imperfect, motivating our discussion of QLIKE loss alongside squared errors.

More recently, Bozovic (2024) examines VIX-managed portfolios and finds that VIX-based timing produces economically significant improvements in portfolio performance, though the mechanism operates through drawdown reduction rather than return enhancement—consistent with our insurance framework.

2.3 The Information Aggregation Argument

We argue that VIX’s forecasting dominance is not accidental but structural. Options markets aggregate information from heterogeneous traders—fundamental analysts who track earnings volatility, macro strategists who monitor the yield curve, systematic traders who exploit cross-asset correlations, and market makers who process order flow. Each of the eleven signal families we test represents one channel of information that at least some option traders monitor. VIX, as the price-weighted consensus of their collective activity, already incorporates these inputs.

This argument generates a testable prediction: signals that are *already in the option-market information set* should show zero incremental forecasting power after controlling for VIX, while signals that are *outside* this set (e.g., truly private information, intraday microstructure) might still add value. Our empirical results are consistent with this prediction.

2.4 Prior Horse Races

The canonical volatility model horse race is [Hansen and Lunde \(2005\)](#), who compare 330 ARCH-type models for forecasting daily exchange rate and IBM stock volatility using the Model Confidence Set (MCS) framework subsequently formalized in [Hansen et al. \(2011\)](#). Their central finding—that no model significantly outperforms the simple GARCH(1,1) when realized volatility is the benchmark—sets the precedent for our study, which extends the horse race from parametric models to signal families and uses VIX rather than GARCH as the benchmark to beat. [Patton and Sheppard \(2015\)](#) decompose realized volatility into good (upside) and bad (downside) components and show that the asymmetric component improves volatility forecasts, a finding we revisit in our variance risk premium tests.

Among signal-specific studies, [Corsi \(2009\)](#) proposes the HAR-RV model using multi-frequency realized volatility components and shows it forecasts well, but does not benchmark against VIX. [Bucci \(2020\)](#) finds that neural networks can improve realized volatility forecasts but does not test whether the improvement survives after conditioning on VIX. [Bollerslev et al. \(2009\)](#) show that VRP predicts equity *returns* (not volatility), but subsequent out-of-sample tests have been mixed.

Our contribution differs from these studies in three ways: (i) we test eleven signal families simultaneously rather than one at a time; (ii) we use a unified evaluation pipeline with consistent lag, cost, and statistical conventions; and (iii) we frame the null as informative rather than disappointing, connecting it to option-market efficiency.

3 Data and the Eleven Signal Families

3.1 Core Data

Our primary dataset consists of daily closing prices for the S&P 500 index (via SPY ETF), the CBOE Volatility Index (VIX), VIX 3-month futures index (VIX3M), gold (GLD ETF), and the 10-year and 3-month U.S. Treasury yields (TNX, IRX). All price data are sourced from Yahoo Finance. The full sample spans January 1993 to March

2026, yielding 8,325 trading days for the VIX–SPY pair. Individual signal families have shorter samples depending on data availability (detailed in Table 1).

We compute 22-day realized volatility as:

$$RV_t^{(22)} = \sqrt{\frac{252}{22} \sum_{i=0}^{21} r_{t-i}^2}, \quad (3)$$

where $r_t = \ln(P_t/P_{t-1})$ is the daily log return. This serves as the forecast target for all volatility prediction exercises.

3.2 The Eleven Signal Families

We pre-specify eleven signal families spanning six broad categories: cross-asset, derivatives-based, behavioral, macroeconomic, alternative data, and calendar. Table 1 summarizes each family.

Table 1: The Eleven Signal Families

#	Signal Family	Category	Construction	Sample	N
1	Cross-asset vol momentum	Cross-asset	5d ΔRV : TLT/USO/UUP/GLD/HYG-LQD	2010–2026	4,053
2	VIX term structure	Derivatives	VIX/VIX3M ratio (contango/backwardation)	2008–2026	4,566
3	Behavioral sentiment	Behavioral	SKEW + VIX/VIX3M + VIX momentum	2011–2026	3,760
4	Variance risk premium	Derivatives	VRP = VIX – 22d realized vol	2006–2026	5,090
5	Multi-asset optimization	Portfolio	MaxDiv/MinVar/RP, 2–9 assets	2023–2026	811
6	Equal risk contribution	Portfolio	Maillard et al. (2010) ERC	2023–2026	811
7	Bitcoin volatility	Cross-asset	BTC 22d RV, BTC–VIX Granger	2015–2026	2,800
8	Yield curve slope	Macro	10Y – 3M Treasury spread	2005–2026	5,337
9	Google Trends fear	Alt. data	5 fear search terms (weekly)	2010–2026	797
10	Overnight VIX	Microstructure	$ VIX_{\text{open}} - VIX_{\text{close}, t-1} $	2010–2026	4,082
11	Calendar anomaly	Calendar	Monthly/seasonal dummies	1993–2026	8,325

Notes: All signals are computed using information available at market close on day $t - 1$ and applied to returns. Google Trends data are weekly and converted to daily by forward-filling. Sample sizes vary by data availability. VIX3M begins in 2008; Bitcoin reliable pricing begins in 2015; Google Trends via pytrends begins in 2004 but we use 2010+ for stability.

3.2.1 Family 1: Cross-Asset Volatility Momentum

Cross-asset volatility spillovers have been documented by [Diebold and Yilmaz \(2012\)](#) and [Barunik et al. \(2016\)](#). We compute the 5-day change in 22-day realized volatility for five asset classes: bonds (TLT), oil (USO), U.S. dollar (UUP), gold (GLD), and credit (HYG minus LQD spread). A composite momentum signal averages the normalized changes. The hypothesis is that volatility shocks propagate across asset classes with a lead-lag structure.

3.2.2 Family 2: VIX Term Structure

The VIX term structure—the ratio of spot VIX to VIX3M—has been proposed as a regime indicator. Persistent contango ($VIX < VIX3M$) is associated with complacency, while backwardation signals acute fear ([Luo and Zhang, 2019](#)). We test whether this ratio adds information beyond VIX level for both volatility prediction and strategy timing.

3.2.3 Family 3: Behavioral Sentiment Index

Investor sentiment affects asset prices through both rational and behavioral channels ([Baker and Wurgler, 2006](#)). We construct a composite Behavioral Sentiment Index (BSI) using four components: the CBOE SKEW index, the VIX/VIX3M ratio, VIX 22-day momentum, and VIX level. Each component is converted to a percentile rank over a 252-day rolling window, and the BSI is the equal-weighted average.

3.2.4 Family 4: Variance Risk Premium

The variance risk premium ($VRP = VIX - RV^{(22)}$) has attracted attention since [Bollerslev et al. \(2009\)](#) showed it predicts equity returns. We distinguish between VRP as a *return* predictor (Bollerslev’s original finding) and VRP as a *volatility* predictor, testing whether VRP carries orthogonal information about future volatility after controlling for VIX level.

3.2.5 Family 5: Multi-Asset Portfolio Optimization

[DeMiguel et al. \(2009\)](#) famously showed that $1/N$ portfolios outperform optimized portfolios out of sample. We test four optimization methods (equal-weight, minimum variance, maximum diversification, risk parity) across seven asset subsets (2 to 9 assets including SPY, GLD, TLT, QQQ, EEM, VWO, BND, DBC, IEFA), evaluating whether expanding the asset universe beyond SPY/GLD improves risk-adjusted returns.

3.2.6 Family 6: Equal Risk Contribution

The Equal Risk Contribution (ERC) portfolio ([Maillard et al., 2010](#)) equalizes marginal risk contributions through constrained optimization. We implement ERC with monthly hold-and-drift rebalancing and test 2-, 3-, and 4-asset variants against the static 50/50 SPY/GLD benchmark.

3.2.7 Family 7: Bitcoin Volatility

Bitcoin has been proposed as a “digital gold” and potential diversifier ([Bouri et al., 2017](#)). We test whether Bitcoin realized volatility contains Granger-causal information for VIX and whether a Bitcoin allocation improves portfolio performance. The *fear channel*—whether crypto stress transmits to traditional markets—is examined through asymmetric Granger causality.

3.2.8 Family 8: Yield Curve Slope

The yield curve slope (10Y – 3M Treasury spread) is a well-known recession predictor ([Estrella and Mishkin, 1998](#)). We test whether curve inversion contains volatility information beyond VIX, exploiting the fact that yield curve movements are slow-moving and potentially capture different risk horizons than VIX.

3.2.9 Family 9: Google Trends Fear Proxy

Internet search behavior provides a real-time measure of retail investor attention and fear ([Da et al., 2011](#); [Preis et al., 2013](#)). We construct a composite Google Fear Index

from the search volume of five terms (“stock market crash,” “recession,” “VIX,” “market crash,” “bear market”), using rolling z -scores over a 52-week window.

3.2.10 Family 10: Overnight VIX Changes

Overnight price changes reflect the processing of after-hours news, earnings announcements, and geopolitical events (Lou et al., 2019). We use the absolute overnight VIX change ($|VIX_{\text{open},t} - VIX_{\text{close},t-1}|$) as a proxy for the information content of overnight news.

3.2.11 Family 11: Calendar Anomaly

The “Halloween effect” and other calendar anomalies have been documented in equity markets (Bouman and Jacobsen, 2002). We test whether VIX-based timing is simply a disguised calendar strategy by examining the correlation between VIX-implied weights and monthly/seasonal dummies.

4 Forecast Design and Benchmark Models

4.1 Unified Forecast Pipeline

All forecast evaluations follow a unified pipeline to ensure comparability:

1. **Signal construction:** Each signal is computed using only information available at the close of day $t - 1$.
2. **Lag enforcement:** The signal is explicitly shifted by one trading day (`signal = signal.shift(1)`) before any use in forecasting or portfolio construction.
3. **Rolling estimation:** Where applicable, parameters are estimated on a rolling window of 252 trading days (approximately one year), re-estimated daily.
4. **Forecast target:** The dependent variable is 22-day forward realized volatility, $RV_{t+1:t+22}$.

4.2 VIX Benchmark

The VIX benchmark model is a simple linear regression:

$$RV_{t+1:t+22} = \alpha + \beta \cdot \text{VIX}_t + \varepsilon_t. \quad (4)$$

This is our primary benchmark. Any candidate signal must demonstrate statistically significant improvement over this specification.

4.3 Incremental Information Test

For each signal family j , we estimate:

$$RV_{t+1:t+22} = \alpha + \beta_1 \cdot \text{VIX}_t + \beta_2 \cdot \text{Signal}_{j,t} + \varepsilon_t, \quad (5)$$

and test $H_0 : \beta_2 = 0$. We report both the in-sample t -statistic for β_2 and the partial correlation $r_{\text{partial}} = \text{cor}(RV, \text{Signal}_j \mid \text{VIX})$.

4.4 Loss Functions and Proxy Robustness

A critical methodological issue in volatility forecast evaluation is that the true conditional variance σ_t^2 is unobservable; any evaluation must use an imperfect proxy. [Patton \(2011\)](#) shows that certain loss functions produce consistent model rankings regardless of the proxy used, while others do not. Specifically, QLIKE and MSE applied to σ^2 (or an unbiased proxy thereof) are *proxy-robust*—they rank models consistently whether the proxy is realized variance from 5-minute returns, daily squared returns r_t^2 , or any other conditionally unbiased estimator.

This result has a subtle but important implication for our evaluation. Daily absolute returns $|r_t|$ are *not* an unbiased proxy for σ_t : the ratio $\mathbb{E}[r_t^2]/\mathbb{E}[|r_t|]^2 = \pi/2 \approx 1.571$ under normality (empirically 2.44 in our sample, reflecting heavy tails), so $|r_t|$ systematically understates σ_t by approximately 19.6%. Forecast evaluations conducted on $|r_t|$ targets can produce different model rankings than evaluations on r_t^2 targets—a finding we confirm

empirically in Section 7.6.

We therefore adopt a three-metric evaluation framework:

1. **QLIKE on r_t^2** (primary): The quasi-likelihood loss $\text{QLIKE} = \ln h_t + r_t^2/h_t$, where h_t is the model’s variance forecast. This is proxy-robust per Patton (2011) and penalizes under-prediction more heavily than over-prediction, consistent with the risk management motivation of volatility forecasting.
2. **Spearman rank correlation** (secondary): The rank correlation between model forecasts and realized r_t^2 , providing a distribution-free measure of monotonic association that is robust to outliers and does not require specifying a loss function.
3. **Out-of-sample R^2** (benchmark comparison): Two variants described below, providing direct comparability with the existing literature.

4.5 Out-of-Sample R^2

We report two complementary measures of out-of-sample forecasting accuracy. The first follows the standard convention of Campbell and Thompson (2008), defining R_{OOS}^2 relative to the prevailing historical mean:

$$R_{\text{OOS,CT}}^2 = 1 - \frac{\sum_{t \in \text{OOS}} (RV_t - \hat{R}V_t^{\text{model}})^2}{\sum_{t \in \text{OOS}} (RV_t - \bar{R}V_{1:t-1})^2}, \quad (6)$$

where $\bar{R}V_{1:t-1}$ is the expanding-window historical mean of realized volatility. Positive values indicate that the model forecasts better than the naive historical average. This metric is directly comparable to the large literature on return and volatility predictability that uses the Campbell-Thompson framework.

The second measure, which is our primary metric, defines R_{OOS}^2 relative to VIX-only forecasts:

$$R_{\text{OOS,VIX}}^2 = 1 - \frac{\sum_{t \in \text{OOS}} (RV_t - \hat{R}V_t^{\text{model}})^2}{\sum_{t \in \text{OOS}} (RV_t - \hat{R}V_t^{\text{VIX}})^2}, \quad (7)$$

where positive values indicate improvement over VIX. This is a more stringent test because VIX is itself a strong predictor (Campbell-Thompson $R_{\text{OOS,CT}}^2$ for VIX alone ranges

from 0.24 to 0.64 across eras). Our research question is specifically whether any signal adds value *beyond* VIX, making the VIX-relative metric the appropriate primary measure. However, reporting both metrics ensures that our results are comparable to the existing literature and demonstrates that VIX’s own forecasting power is substantial—many signals that achieve positive $R_{\text{OOS,CT}}^2$ (i.e., beat the historical mean) still show zero $R_{\text{OOS,VIX}}^2$ (i.e., add nothing beyond VIX).

5 Statistical Evaluation and Multiple-Testing Control

5.1 Diebold-Mariano Test

We use the [Diebold and Mariano \(1995\)](#) test to compare forecast accuracy, with squared forecast errors as the loss function:

$$d_t = e_{t,\text{VIX}}^2 - e_{t,\text{model}}^2, \quad (8)$$

where $e_{t,k}$ denotes the forecast error from model k . The DM statistic is:

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})/T}}, \quad (9)$$

with [Newey and West \(1987\)](#) heteroskedasticity and autocorrelation consistent (HAC) standard errors to account for serial correlation in multi-step-ahead forecasts (bandwidth = 22, matching the forecast horizon).

5.2 Multiple-Testing Corrections

When many predictors are tested, conventional $|t| > 1.96$ thresholds produce excessive false discoveries. [Harvey et al. \(2016\)](#) address this problem in the context of asset pricing, presenting a framework in which the appropriate significance threshold depends on the number of tested strategies and the assumed prior probability of a true discovery.

Their analysis—applied to roughly 300 factors in the cross-section literature—yields an approximate threshold of $|t| > 3.0$ under conservative priors. We adopt $|t| > 3.0$ as an approximate conservative threshold motivated by their analysis, noting that for our 11 tests the simple Bonferroni correction yields a comparable threshold of $|t| > 3.23$ (using $\alpha = 0.05/11$).

To provide formal multiple-testing control beyond this rule of thumb, we implement the Holm-Bonferroni sequential procedure (Holm, 1979) across the 11 primary Diebold-Mariano tests. The procedure works as follows: (i) rank the 11 raw p -values from smallest ($p_{(1)}$) to largest ($p_{(11)}$); (ii) for rank k , reject H_0 if $p_{(k)} \leq \alpha/(11 - k + 1)$; (iii) stop at the first non-rejection. We report both raw and Holm-Bonferroni adjusted p -values for all tests. This procedure controls the family-wise error rate (FWER) at $\alpha = 0.05$ while being strictly more powerful than Bonferroni. As we show in Section 7, the Holm-Bonferroni correction does not change any conclusion—all 11 signals fail both the raw and corrected tests—which strengthens the robustness of our null finding.

5.3 Model Confidence Set

Beyond pairwise DM tests, we employ the Model Confidence Set (MCS) procedure of Hansen et al. (2011) to identify the set of models that cannot be statistically distinguished from the best model at a given confidence level. The MCS procedure has three advantages over pairwise testing: (i) it controls for the joint testing problem without requiring a pre-specified benchmark; (ii) it produces a *set* of superior models rather than a single winner, honestly reflecting the uncertainty in model ranking; and (iii) it accounts for the dependence structure among forecast errors through block bootstrap resampling (block length = 22, matching the forecast horizon).

Formally, let $\mathcal{M}_0$ denote the full set of candidate models. At each step, the MCS procedure tests the null hypothesis that all surviving models have equal expected loss. If rejected, the worst-performing model is eliminated. The procedure terminates when the null cannot be rejected at significance level α , yielding the model confidence set $\hat{\mathcal{M}}_\alpha^*$. We report MCS results at $\alpha = 0.10$.

We apply the MCS across our volatility model families (GJR-GARCH, AMEM, MEM, GARCH, HAR, EWMA) using QLIKE on r_t^2 as the loss function. As shown in Section 7.6, the MCS membership is criterion-dependent—different loss functions can produce different confidence sets—which itself strengthens the VIX sufficiency argument.

5.4 Cross-Era Validation

To prevent any single market regime from driving conclusions, we partition the sample into five non-overlapping eras:

1. **Dot-Com** (1993–2000): Rising equity markets, Asian crisis, LTCM
2. **Post-Dot-Com** (2000–2007): Tech bust, recovery, housing boom
3. **GFC** (2007–2012): Global financial crisis, European debt crisis
4. **Low-Vol QE** (2012–2020): Quantitative easing, low volatility regime
5. **COVID/Inflation** (2020–2026): Pandemic shock, inflation, rate hikes

A finding is considered robust if it holds in at least 4 of 5 eras.

6 Volatility-Timing Strategy Design

6.1 The 12/VIX Strategy

Our primary VT strategy is the 12/VIX rule:

$$w_t^{\text{SPY}} = \min\left(\frac{12}{\text{VIX}_{t-1}}, 1\right), \quad (10)$$

where the equity weight is capped at 100% and the complement $(1 - w_t^{\text{SPY}})$ is allocated to GLD (or cash in pre-GLD eras). This strategy has several desirable properties:

- **Zero parameters:** The numerator 12 approximates long-run average VIX.

- **Smooth weights:** The concave $1/x$ mapping produces gradual weight changes, limiting turnover (mean daily $|\Delta w| = 0.035$, annual turnover $\approx 8.9\times$).
- **Lag-robust:** Because weights change slowly, the 1-day lag between signal and execution has negligible impact.
- **Crowding-resistant:** Market impact analysis shows that even \$10 billion in AUM would reduce Sharpe by less than 60 basis points, and the VIX feedback loop is damping (converging) rather than amplifying.

6.2 Baseline: 50/50 SPY/GLD

The baseline is a static 50/50 allocation to SPY and GLD with monthly rebalancing. This allocation is approximately optimal by construction: SPY annualized volatility (19.3%) $\approx$ GLD annualized volatility (18.3%), so the 50/50 split closely approximates the risk parity solution (Maillard et al., 2010).

6.3 Transaction Costs

We deduct 5 basis points per leg for each weight change:

$$\text{TX}_t = 0.0005 \times (|w_t^{\text{SPY}} - w_{t-1}^{\text{SPY}}| + |w_t^{\text{GLD}} - w_{t-1}^{\text{GLD}}|). \quad (11)$$

Net portfolio return is $r_t^{\text{net}} = r_t^{\text{gross}} - \text{TX}_t$.

7 Main Results and Robustness

7.1 Volatility Forecasting: No Signal Beats VIX

Table 2 presents the main results for all eleven signal families. The key finding is unanimous: no signal produces a statistically significant out-of-sample improvement over VIX.

Table 2: Volatility Forecasting: Eleven Signal Families vs. VIX

#	Signal Family	Partial $r VIX$	IS ΔR^2	IS t -stat	$R^2_{OOS,CT}$	$R^2_{OOS,VIX}$	DM $ t $	Raw p	HB p
1	Cross-asset vol momentum	0.087	−0.022	−1.45	0.312	−0.022	1.45	0.147	1.000
2	VIX term structure	0.181	0.033	17.6***	0.398	0.010	0.87	0.384	1.000
3	Behavioral sentiment	0.091	0.004	1.64	0.365	< 0.001	0.52	0.603	1.000
4	Variance risk premium	0.054	< 0.003	3.51**	0.381	< 0.003	1.21	0.226	1.000
5	Multi-asset optimization	—	—	—	—	—	—	—	—
6	Equal risk contribution	—	—	—	—	—	—	—	—
7	Bitcoin volatility	0.178	—	—	—	—	—	—	—
8	Yield curve slope	< 0.08	0.009	2.51	0.370	< 0.001	0.84	0.401	1.000
9	Google Trends fear	0.271	0.038	7.92***	0.361	< 0.001	0.67	0.503	1.000
10	Overnight VIX	0.030	0.005	5.07***	0.374	0.003	1.12	0.263	1.000
11	Calendar anomaly	−0.007	0.012	−2.39*	0.348	0.000	0.15	0.881	1.000
<i>VIX-only benchmark</i>		—	—	—	0.365	—	—	—	—
<i>Harvey (2016) approx. threshold</i>							> 3.0		
<i>Holm-Bonferroni FWER</i>									$\alpha = 0.05$

Notes: Partial $r|VIX$ is the partial correlation between the signal and 22-day forward realized volatility, controlling for the VIX level. IS ΔR^2 is the in-sample increase in R^2 from adding the signal to a VIX-only regression. $H_0 : \beta_{\text{signal}} = 0$. $R^2_{OOS,CT}$ is the [Campbell and Thompson \(2008\)](#) out-of-sample R^2 relative to the expansion of the historical mean (Equation 6). $R^2_{OOS,VIX}$ is the out-of-sample R^2 relative to VIX-only forecasts (Equation 6). $|t|$ is the absolute Diebold-Mariano statistic with [Newey and West \(1987\)](#) HAC standard errors (bandwidth = 2). p is the two-sided p -value from the DM test. HB p is the Holm-Bonferroni adjusted p -value controlling FWER across the 8 testable families. Families 5–6 are portfolio strategies without direct volatility forecasting. Family 7 (Bitcoin) is tested for Granger causality rather than regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$ (only, conventional thresholds).

All signals lagged one trading day. Note that several signals achieve positive $R^2_{OOS,CT}$ (i.e., they beat the historical mean) but zero or negative $R^2_{OOS,VIX}$ (i.e., they add nothing beyond VIX), confirming that VIX itself is a strong predictor.

Several patterns are noteworthy. First, the dual R^2 metrics reveal a striking pattern: most signal families achieve positive Campbell-Thompson $R^2_{OOS,CT}$ values in the range of 0.31–0.40, indicating that they do forecast volatility better than the naive historical mean. However, these same signals show zero or negative VIX-relative $R^2_{OOS,VIX}$, confirming that their forecasting power is entirely subsumed by VIX. The VIX-only benchmark itself

achieves $R_{\text{OOS,CT}}^2 = 0.365$, which accounts for almost all of the predictive content in every signal family.

Second, the Holm-Bonferroni correction reinforces the null. All eight testable DM tests produce raw p -values well above 0.05 (minimum raw $p = 0.147$ for cross-asset momentum), and the adjusted p -values are uniformly 1.000. The multiplicity correction is strictly redundant here—no individual test even approaches marginal significance—but reporting it demonstrates that our conclusions are robust to formal multiple-testing control.

Third, there is a stark disconnect between in-sample and out-of-sample performance. Google Trends fear (Family 9) achieves a partial correlation of 0.271 and an in-sample ΔR^2 of 0.038 with a highly significant t -statistic of 7.92, yet delivers essentially zero out-of-sample improvement (DM $|t| = 0.67$, raw $p = 0.503$). This is consistent with overfitting—the fear signal captures contemporaneous co-movement with VIX but has no genuine predictive lead.

Second, the variance risk premium (Family 4) illustrates the distinction between return prediction and volatility prediction. [Bollerslev et al. \(2009\)](#) showed that VRP predicts equity *returns* ($t = 3.51$ in our in-sample test, consistent with their finding). But VRP does not predict *volatility* beyond VIX—unsurprisingly, since VRP is defined as VIX minus realized volatility, making it mechanically correlated with VIX.

Third, the overnight VIX change (Family 10) shows the most promising incremental R^2 (+0.45%), but the improvement vanishes after transaction costs. The overnight signal generates high-frequency weight changes that are too expensive to exploit.

7.2 Strategy Comparison: No Signal-Based Strategy Beats Benchmarks

Table 3 compares the trading strategies derived from each signal family against the 12/VIX and 50/50 SPY/GLD benchmarks over the common evaluation period.

Table 3: Volatility-Timing Strategies: Signal Families vs. Benchmarks

#	Strategy	Sharpe	Δ Sharpe	MDD (%)	Turnover	DM $ t $
<i>Benchmarks</i>						
—	Buy-and-hold 50/50 SPY/GLD	0.947	—	−32.5	0.0	—
—	12/VIX (SPY/GLD)	0.870	−0.077	−32.3	8.9×	—
<i>Signal-based strategies</i>						
1	Cross-asset composite	0.810	−0.060	−34.1	11.2×	0.93
2	VIX term structure (contango boost)	0.880	+0.010	−31.8	9.4×	0.31
3	Behavioral sentiment (fear hedge)	0.900	+0.030	−30.2	10.1×	0.72
4	VRP timing	0.782	−0.088	−30.8	12.3×	0.65
4	VRP percentile	0.800	−0.070	−30.5	11.8×	0.58
6	ERC (2-asset)	0.795	−0.054	−13.3	4.2×	1.02
8	Yield curve + VIX combo	0.746	−0.124	−35.1	9.8×	1.15
9	Google Trends fear	0.834	−0.036	−33.7	10.5×	0.48
10	Overnight VIX timing	0.851	−0.019	−31.9	14.7×	0.37
11	Calendar-only	0.658	−0.212	−48.4	2.0×	1.89

Notes: Sharpe ratios computed over the longest common sample for each pair. Δ Sharpe is relative to 12/VIX. MDD is maximum drawdown. Turnover is annualized two-way. DM $|t|$ tests whether the signal-based strategy significantly outperforms 12/VIX. All strategies use 5 bps per leg transaction costs and `signal.shift(1)` lag. Harvey threshold $|t| > 3.0$ required; none passes.

The most striking result is that the static 50/50 SPY/GLD allocation (Sharpe = 0.947) outperforms *every* dynamic strategy, including the 12/VIX benchmark (Sharpe = 0.870). This confirms our earlier finding that VT strategies sacrifice return in exchange for drawdown protection.

Among signal-based strategies, the VIX term structure “contango boost” variant comes closest to matching 12/VIX, with a Sharpe improvement of +0.010, but this is far from statistical significance (DM $|t|$ = 0.31). The behavioral sentiment strategy

achieves Sharpe 0.900 but does not significantly outperform 12/VIX (DM $|t| = 0.72$).

7.3 Multi-Asset Portfolio Optimization

Table 4 presents the multi-asset portfolio optimization results from our systematic evaluation of four methods across seven asset subsets.

Table 4: Multi-Asset Portfolio Optimization: 50/50 SPY/GLD vs. Alternatives

Method	N_{assets}	Sharpe	MDD (%)	Beats 50/50?
50/50 SPY/GLD (baseline)	2	1.849	−13.3	—
Equal Weight	3	1.492	−11.2	No
Equal Weight	9	1.285	−8.3	No
Min Variance	3	1.412	−10.8	No
Min Variance	9	1.308	−5.7	No
Max Diversification	3	1.385	−10.5	No
Risk Parity	3	1.401	−10.9	No
Inverse Volatility	2	1.795	−13.3	No
ERC	2	1.795	−13.3	No

Notes: Common evaluation period 2023-01-04 to 2026-03-27 (811 trading days). All methods use monthly hold-and-drift rebalancing, 5 bps per leg TX costs, `signal.shift(1)`. 9-asset universe: SPY, GLD, TLT, QQQ, EEM, VWO, BND, DBC, IEFA. More assets reduce MDD but also reduce Sharpe. The 2-asset ERC and Inverse Volatility converge to approximately 50/50 because SPY vol (19.3%) $\approx$ GLD vol (18.3%).

The result is unambiguous: no multi-asset strategy beats the simple 50/50 SPY/GLD allocation on Sharpe ratio. Adding assets reduces maximum drawdown (9-asset minimum variance achieves MDD of only −5.7% versus −13.3% for 50/50) but at the cost of lower returns, resulting in lower Sharpe ratios. This is consistent with [DeMiguel et al. \(2009\)](#), who show that estimation error in optimized portfolios offsets diversification benefits.

We further test whether financial-stress alt-data (NFCI, STLFSI, USEPU, WLEMU) can improve a daily regime-switching allocation between SPY and GLD. The matched-pair design (K1123, $N = 1,819$ trading days, 2019–2026) tests five alt-data allocation rules against the static 50/50 baseline (Sharpe = 1.31, OOS Sharpe = 1.98). All five alt-data strategies underperform the baseline on full-sample Sharpe (best: 1.16) and on out-of-sample Sharpe. This extends the allocation null from the portfolio-optimization domain (Table 4) to the regime-switching domain: VIX sufficiency holds regardless of whether the application is volatility forecasting, volatility-timing strategy, multi-asset optimization, or alt-data regime allocation.

7.4 Era Stability of VIX Sufficiency

Table 5 presents the VIX forecasting power and strategy performance across five non-overlapping eras.

Table 5: VIX Sufficiency Across Five Market Eras (1993–2026)

Era	Period	N	VIX–RV R^2	β	t -stat	Mean VIX	Mean RV
1. Dot-Com	1993–2000	1,811	0.525	0.810	44.7	18.5	14.6
2. Post-Dot-Com	2000–2007	1,820	0.645	0.868	57.4	19.3	15.7
3. GFC	2007–2012	1,261	0.508	0.957	36.1	26.3	22.4
4. Low-Vol QE	2012–2020	1,908	0.244	0.692	24.8	15.0	11.8
5. COVID/Inflation	2020–2026	1,525	0.309	0.810	26.1	21.0	17.2
Full Sample	1993–2026	8,325	0.514	0.879	93.8	19.5	15.8
Cross-era statistics:			Mean $R^2 = 0.446$		CV = 0.37 (time-invariant)		

Notes: R^2 from regressing 22-day forward realized volatility on VIX level. β is the slope coefficient. All t -statistics are highly significant ($p \approx 0$). The lowest R^2 (0.244) occurs during the QE era when VIX was compressed near 15 with low variation; the highest (0.645) occurs during the post-dot-com era with wider VIX range. CV = coefficient of variation of R^2 across eras; CV < 0.50 indicates time-invariance.

The VIX–RV relationship is remarkably stable. The coefficient of variation of R^2 across eras is 0.37—well below the 0.50 threshold we adopt for time-invariance. Importantly, the low R^2 in the QE era (0.244) does not indicate VIX failure; rather, it reflects the mechanical compression of both VIX and RV during this period (VIX stuck near 15, RV near 12), which reduces regression R^2 without impairing VIX’s relative forecasting accuracy.

7.5 Competing Signals by Era

Table 6 presents the incremental R^2 of three representative competing signals across the five eras.

Table 6: Incremental R^2 of Competing Signals Across Eras

Signal	Era 1	Era 2	Era 3	Era 4	Era 5	Harvey Pass?
Overnight VIX (abs)	0.0002	0.0001	0.0039 [†]	0.0006	0.0032	1/5
VRP proxy	0.0005	0.0002	0.0160 [†]	0.0008	0.0005	1/5
Vol momentum 20/60	0.0001	0.0004	0.0216 [†]	0.0001	0.0372 [†]	2/5

Notes: Incremental R^2 from adding the signal to a VIX-only regression of 22-day forward realized volatility. Values are near zero in non-crisis eras (Era 1 DotCom, Era 2 PostDotCom, Era 4 LowVol/QE), confirming VIX sufficiency in normal regimes. [†]Denotes an in-sample Harvey $|t| > 3.0$ threshold crossing in the indicated era; four such crossings occur—three in Era 3 (GFC: all three signals) and one in Era 5 (COVID/Inflation: Vol momentum 20/60). These in-sample crisis-era exceptions do not survive out-of-sample Diebold–Mariano testing with Holm–Bonferroni correction (Table ??), consistent with the paper’s core VIX-sufficiency finding on average while identifying well-defined crisis-regime boundary conditions.

The pattern is nuanced: during the three non-crisis eras (DotCom, PostDotCom, LowVol/QE), incremental R^2 values are uniformly tiny (all below 0.001) across all signals, confirming VIX sufficiency in normal market regimes. However, during the GFC (Era 3)

and COVID/Inflation (Era 5) eras, three signals—Overnight VIX absolute change, VRP proxy, and Vol momentum 20/60—exhibit materially larger incremental R^2 values (up to 0.037 for Vol momentum in Era 5), with four in-sample cells crossing the Harvey threshold: all three signals in Era 3 GFC, and Vol momentum 20/60 in Era 5. This in-sample crisis-era exception is consistent with the paper’s core out-of-sample finding, however: all era-pooled OOS Diebold–Mariano tests remain $|t| < 3$ even after Holm–Bonferroni correction (Table ??). The refined reading is that VIX sufficiency is robust on average but has identified, well-defined crisis-regime boundary conditions where the option-market information set may be partially incomplete during extreme volatility episodes.

A complementary matched-pair design directly tests whether alt-data signals improve forecasting specifically during VIX jump days (K1117, $N_{\text{jump}} = 181$ days at the VIX 2σ threshold, $N_{\text{total}} = 4,022$). Six alt-data variables (VVIX, USEPU, NFCI, ANFCI, STLFSI, WLEMU) are tested for incremental forecasting content on jump-day windows versus control windows. The verdict is FULL_NULL: no signal achieves DM-HLN $|t| > 2$ in any specification, and no hypothesis (H_1 , H_2 , H_3) passes BH correction (VVIX: H_1 DM $t = 1.35$, $p_{\text{BH}} = 0.69$). VIX sufficiency extends even to extreme volatility episodes where option-market information updating is fastest.

7.6 Criterion-Dependent Model Rankings

An important robustness finding emerges when we evaluate forecast models under the [Patton \(2011\)](#) framework. Table 7 compares model rankings across three evaluation criteria—QLIKE on r_t^2 , Spearman rank correlation, and sub-period stability—for six volatility model families estimated on the full 2008–2026 out-of-sample period ($N = 4,589$).

Table 7: Criterion-Dependent Model Rankings (2008–2026, $N = 4,589$)

	GJR	AMEM	MEM	GARCH	HAR	EWMA
<i>Panel A: QLIKE on r_t^2 (proxy-robust)</i>						
QLIKE	1.527	1.559	1.576	1.576	1.649	1.624
Rank	1	2	3	4	6	5
<i>Panel B: Spearman rank correlation with r_t^2</i>						
ρ_S	0.418	0.398	0.376	0.373	0.362	0.356
Rank	1	2	3	4	5	6
<i>Panel C: MCS membership ($\alpha = 0.10$)</i>						
In MCS?	✓	✗	✗	✗	✗	✗
<i>Panel D: Pairwise DM tests (QLIKE, HAC bandwidth = 22)</i>						
GJR vs. AMEM	DM = 3.78, $p < 0.001$; GJR $\succ$ AMEM (Harvey et al. (2016) pass)					
GJR vs. GARCH	DM = 4.76, $p < 0.001$; GJR $\succ$ GARCH (Harvey et al. (2016) pass)					
AMEM vs. GARCH	DM = 2.85, $p = 0.004$; AMEM $\succ$ GARCH (marginal)					

Notes: All models estimated with expanding window (minimum 500 observations, refit every 63 days). QLIKE on r_t^2 is proxy-robust per Patton (2011). MCS uses block bootstrap ($B = 5,000$, block length = 22) per Hansen et al. (2011). GJR-GARCH dominates across all criteria, with the MCS containing only GJR at the 10% level. Note: under $|r_t|$ targets (not shown), AMEM outperforms GJR on QLIKE—criterion choice reverses the ranking for non-robust loss functions, confirming the importance of the Patton framework.

Three findings emerge. First, GJR-GARCH is the dominant model under the proxy-robust QLIKE criterion, with DM = 3.78 ($p < 0.001$) versus the next-best AMEM—passing the Harvey et al. (2016) threshold. The MCS at $\alpha = 0.10$ contains only GJR, indicating statistical separation from all competitors.

Second, the rankings are *criterion-dependent*: under $|r_t|$ targets (which are *not* proxy-robust), AMEM outperforms GJR on QLIKE, reversing the ranking. This reversal occurs because $|r_t|$ introduces a 19.6% downward bias in the volatility proxy (empirical ratio

$\mathbb{E}[r_t^2]/\mathbb{E}[|r_t|]^2 = 2.44$ versus the normal-theory $\pi/2 = 1.57$), which differentially affects models with asymmetric specifications. The Patton framework resolves this ambiguity by privileging r_t^2 -based evaluation.

Third—and most relevant for VIX sufficiency—this criterion dependence *strengthens* rather than weakens our main finding. Even among sophisticated volatility models, the “best” model depends on which evaluation criterion is used. Yet none of them, regardless of criterion, improves upon VIX as an input signal. The instability of model rankings across criteria makes the stability of VIX sufficiency all the more striking.

7.7 12/VIX Strategy by Era

Table 8 shows the 12/VIX strategy performance across eras.

Table 8: 12/VIX Strategy vs. 50/50 Baseline Across Eras

Era	VT Sharpe	BH Sharpe	Δ Sharpe	VT MDD	BH MDD	Winner
1. Dot-Com ^a	1.236	1.261	−0.026	−9.2%	−9.9%	50/50
2. Post-Dot-Com ^a	0.128	0.179	−0.051	−31.3%	−26.3%	50/50
3. GFC	0.537	0.669	−0.132	−32.3%	−32.5%	50/50
4. Low-Vol QE	1.116	0.797	+0.319	−12.8%	−13.1%	12/VIX
5. COVID/Inflation	0.963	1.139	−0.176	−20.2%	−20.3%	50/50
12/VIX wins:			1 of 5 eras			

Notes: ^aGLD not available; complement asset is cash (risk-free rate). All strategies use signal.shift(1) and 5 bps per leg TX costs. 12/VIX wins only in the low-volatility QE era when persistent low VIX enables consistent high equity exposure (mean weight $\approx 80\%$).

The 12/VIX strategy wins in only 1 of 5 eras (the low-volatility QE period). This further reinforces that VT strategies do not generate alpha in the traditional sense; their value lies in drawdown insurance, which is most apparent during high-volatility regimes where VT reduces exposure.

7.8 Cross-Asset Native-IV Sufficiency: A Seven-Asset PIT Panorama

The VIX-sufficiency claim gains its strongest cross-asset confirmation from a seven-asset Publication-In-Time (PIT) panorama. Extending the K1116c (SPY), K1116f (GLD, TLT, BTC-USD), K1201 (QQQ, USO), and K1203 (EEM) experimental series to a unified evaluation, we test four alt-data specifications—**base** (AR(1) + native IV), **epu** (+ EPU uncertainty indices), **finstress** (+ NFCI / ANFCI / STLFSI financial-stress series), and **all** (all alt-data simultaneously)—against a native-IV baseline using PIT-aligned DM-HLN tests (`pit_shift0` primary specification, 2023-01-01 to 2026-04-10, $N_{\text{OOS}} = 170$ weeks).

PIT alignment ensures each weekly regression uses only alt-data releases available at the time of prediction. For FRED weekly series (NFCI, ANFCI, STLFSI), the publication lag is Wednesday of the following week; for EPU indices, the next business day. All specifications include an AR(1) term and the asset’s native implied-volatility proxy as baseline regressors, with the native-IV proxy matched to each asset class: $\hat{\text{VIX}}$ for SPY, QQQ, and EEM;² GVZ for GLD; MOVE for TLT; BVIN for BTC-USD; and OVX for USO.

Table 9 reports the DM-HLN t -statistics for the `pit_shift0` variant across all 28 specification cells ($7 \text{ assets} \times 4 \text{ specifications}$). A positive t indicates the alt-data challenger beats the native-IV baseline on QLIKE; a negative t indicates the baseline wins. The Harvey (2016) threshold of $|t| > 3.0$ is required to declare a statistically significant improvement.

²For EEM, the CBOE Emerging Markets VIX ($\hat{\text{VXEEM}}$) is the natural native-IV proxy, but it was delisted prior to the K1203 evaluation period (`yfinance` returns HTTP 404 for $\hat{\text{VXEEM}}$ and reports “possibly delisted” for VXEEM; $\hat{\text{VXFXI}}$ and $\hat{\text{CIV}}$ are also unavailable). Accordingly, K1203 uses $\hat{\text{VIX}}$ as a systemic spillover proxy (empirical weekly EEM–VIX correlation ≈ 0.75 in this sample) and additionally confirms the verdict against a 30-day EEM realized-volatility (`rv30`) robustness baseline. Both configurations yield a clean NULL, ruling out sensitivity to the IV-proxy choice.

Table 9: Seven-Asset Cross-Asset Panorama: DM-HLN t -Statistics vs. Native-IV Baseline (PIT-Aligned, `pit_shift0`)

Asset	Source	base	epu	finstress	all
SPY	K1116c	−3.021	−2.603	−3.001	−2.537
GLD	K1116f	−2.103	−2.069	−3.341	−2.246
TLT	K1116f	+1.433	−2.477	+3.743 [†]	−5.666
BTC-USD	K1116f	−5.494	−3.550	+1.370	+0.203
QQQ	K1201	−2.186	−1.967	−2.439	−1.967
USO	K1201	−3.049	−5.596	−2.584	−3.735
EEM	K1203 ($\hat{VIX}$)	−2.596	−3.539	−1.434	−0.999

Notes: Positive t indicates the challenger specification beats the native-IV baseline on QLIKE; negative t indicates the baseline wins. Harvey (2016) threshold: $|t| > 3.0$. Negative values exceeding -3.0 in absolute value denote statistically significant *baseline wins*. [†]TLT $\times$ **finstress** ($t = +3.743$) is lag-sensitive: it collapses to $t = +2.000$ under `pit_shift1` (one additional lag week) and the QLIKE improvement is only $+0.50\%$ (below the 5% materiality gate).³ Weekly OOS period 2023-01-01 to 2026-04-10 ($N_{\text{OOS}} = 170$ weeks). EEM native-IV proxy: $\hat{VIX}$ (see footnote on $\hat{VXEEM}$ delisting); `rv30` robustness yields the same qualitative verdict.

Across all 28 panorama cells, only one—TLT $\times$ **finstress** ($t = +3.743$)—crosses the Harvey threshold on the primary specification, and this sole exceedance fails every robustness check: it collapses under one additional lag week and is QLIKE-insignificant. The remaining 27 cells are either null (positive but below $+3.0$) or baseline-wins (negative). The cross-asset verdict is thus **UNIVERSAL_NULL_7/7**: for six of seven assets the null is unambiguous; for TLT the single outlier is a regime artefact rather than a structural signal. Native-IV sufficiency extends from US equity (SPY) to international equity (EEM, QQQ), commodities (GLD, USO), fixed income (TLT), and cryptocurrency

(BTC-USD), constituting the broadest cross-asset evidence for option-market information efficiency available at daily frequency.

7.9 Channel-Specific Mechanism Heterogeneity

The aggregate null across eleven signal families conceals a structured heterogeneity in *how* different model architectures fail. We characterize three channels with qualitatively distinct failure patterns, providing sharper guidance for future research beyond "all signals fail."

7.9.1 Channel 1: HAR+VIX on Equity—Regime-Invariant Gains on Parkinson Volatility

The HAR-RV+VIX model (M4: Corsi 2009 HAR on log-Parkinson realized volatility with $\log(\text{VIX}_{t-1}^2)$ regressor) delivers statistically significant out-of-sample improvements on large-cap and mid-cap equity indices relative to a GJR-GARCH(1,1) Normal baseline evaluated on the same Parkinson target. We define regime-invariance as passing the Harvey threshold ($|t| > 3.0$, BH-adjusted) in all three VIX quintile regimes (low/mid/high).

Across 18 equity asset $\times$ regime cells (3 assets $\times$ 3 regimes $\times$ 2 models, evaluated in K1137), HAR+VIX achieves regime-invariant status for two equity indices: QQQ passes across all three regimes (low $t = 6.79$, mid $t = 4.83$, high $t = 3.09$; all BH-adjusted $p < 0.001$), and IWM passes across all three regimes (low $t = 4.22$, mid $t = 3.61$, high $t = 2.73$; all BH-adjusted $p < 0.05$). SPY achieves 2 of 3 regime passes (low $t = 7.99$, mid $t = 4.39$, but high $t = 2.26$ fails BH correction). Across the full 54-cell evaluation (6 assets $\times$ 3 regimes $\times$ 3 models), 17 cells pass BH-adjusted Harvey threshold.

These gains are specific to the Parkinson-range volatility measurement framework.⁴ The HAR+VIX gains reflect the empirically documented efficiency of Parkinson-range estimators for capturing intraday volatility dynamics that daily closing-price GARCH

⁴K1136 fair-test analysis establishes that GARCH-MIDAS-X, a score-driven long-memory extension, achieves zero conditional passes (0/3 regimes) under the same matched-target protocol (K1136, `fair_test_1_m3_midas_pass_count = 0`), ruling out the possibility that any long-run factor model outperforms.

imperfectly approximates—not an incremental signal beyond VIX in the traditional regression sense.

7.9.2 Channel 2: GARCH-MIDAS Long-Run Factor—Universal Null

GARCH-MIDAS-X, which extends the baseline GJR-GARCH by adding a slow-moving multiplicative component driven by lagged monthly VIX-squared ($\tau_t = \exp(m + \theta \times \text{VIX}_{\text{monthly}}^2)$), achieves zero BH-adjusted conditional passes across all three VIX regimes and all six assets evaluated in K1137 (`n_midas_conditional_pass` = 0). This null is universal: neither equity nor commodity assets benefit from the MIDAS long-run factor once target-consistent evaluation is applied.

The MIDAS null strengthens the option-market information aggregation argument: any low-frequency volatility information that a one-month lagged VIX factor could supply is already impounded in the contemporaneous VIX level, leaving no incremental forecasting content for the long-run MIDAS component.

7.9.3 Channel 3: GAS-t Score-Driven Models—Asset-Class Bifurcation

The Generalized Autoregressive Score (GAS-t) framework (Creal et al., 2013) exhibits qualitatively different behavior across asset classes.

Equity: architectural incompatibility. On SPY (K1138: $N_{\text{OOS}} = 1,323$, 2021–2026), GAS-t is actively harmful relative to GJR-GARCH Normal (DM-HLN $t = -3.27$, $p < 0.001$). K1143 tests four candidate remedies—Hansen skew-t asymmetry (M2), score winsorization at ± 0.3 (M3), persistence capping at $\beta \leq 0.9$ (M4), and regime switching to HAR in low-VIX environments (M5)—and finds that *all variants produce QLIKE above the GJR-N baseline*: M1 symmetric GAS-t QLIKE = 1.527 versus GJR-N = 1.474; Hansen skew-t (M2) QLIKE = 1.535 (worse than M1). The post-2020 equity return distribution (excess kurtosis = 8.85, skewness = +0.30) contradicts the GAS framework’s core premise that score-driven updates stabilize conditional variance when innovations are approximately Student- t . Instead, the GAS score exhibits positive skewness (+1.58), causing right-biased updates that systematically overweight large positive return shocks.

Commodity: tail-risk instrument, not volatility forecaster. On negatively skewed commodities (USO, UNG), the skew-t GAS extension (Hansen, 1994) recovers VaR/ES performance that the symmetric GAS-t variant lacks (K1135, H_3 : ES backtests pass at the 1% level for all four assets, BH-adjusted). However, forecasting QLIKE remains null (0 of 4 assets improve, H_1 pass count = 0). This bifurcation—ES improvement without QLIKE improvement—indicates that skew-t GAS captures distributional shape for risk management without improving point volatility forecasts.

The Channel 3 pattern thus identifies a *usage boundary*: for equity volatility forecasting, GAS-t is architecturally incompatible and should be replaced by GJR-GARCH. For commodity tail-risk estimation specifically, skew-t GAS provides a validated tail-risk instrument despite null forecasting performance. These distinctions cannot be inferred from the aggregate null result alone.

Table 10: Channel-Specific Outcome Matrix

Channel	Equity (SPY/QQQ/IWM)	Commodity (USO/GLD)	TLT
1. HAR+VIX (Parkinson target)	PASS (QQQ/IWM 3/3 regimes)	PASS (TLT/USO)	PASS (3/3)
2. GARCH-MIDAS long-run	NULL (0/3 regimes all assets)	NULL	NULL
3a. GAS-t (QLIKE)	HARMFUL (SPY $t = -3.27$)	NULL	—
3b. Skew-t GAS (ES 1%)	—	PASS (4/4 assets)	—

Notes: Channel 1 PASS requires BH-adjusted Harvey $|t| > 3.0$ in all three VIX regimes. Channel 2 GARCH-MIDAS-X fair test (matched target). Channel 3a tests QLIKE improvement vs GJR-GARCH. Channel 3b tests ES backtest at 1% coverage. Regime-invariant = PASS in all low/mid/high VIX regimes.

8 Why the Null is Informative

8.1 The Information Aggregation Explanation

Our null result is not a failure to find signal but a positive finding about market efficiency. The VIX option market is one of the deepest and most liquid derivatives markets in the world, with average daily trading volume exceeding 500,000 contracts. This liquidity

ensures rapid price discovery, meaning that any signal with predictive content for short-term volatility is quickly impounded into option prices and hence into VIX.

We can formalize this argument. Let Ω_t denote the information set of the representative option trader at time t . Under semi-strong form efficiency of the options market:

$$\mathbb{E}[\sigma_{t+1:t+22}^2 | \Omega_t] = f(\text{VIX}_t), \quad (12)$$

where $f(\cdot)$ is monotonically increasing. Any signal $s_t \in \Omega_t$ satisfies:

$$\mathbb{E}[\sigma_{t+1:t+22}^2 | \text{VIX}_t, s_t] = \mathbb{E}[\sigma_{t+1:t+22}^2 | \text{VIX}_t], \quad (13)$$

i.e., VIX is a sufficient statistic for all information already incorporated in option prices. Our eleven signal families are all elements of Ω_t , which explains the zero incremental forecasting power.

8.2 What Might Break VIX Sufficiency?

The sufficiency argument identifies two conditions under which alternative signals could add value:

1. **Higher-frequency information:** Intraday 5-minute realized volatility *may* capture microstructure dynamics (e.g., opening auctions, lunch-hour liquidity, closing imbalances) that closing-price VIX does not fully reflect. The direct evidence from our K745 pilot ($N = 37$), however, runs in the opposite direction: daily HAR-ABS (QLIKE = 0.077) outperformed 5-minute HAR-RV (QLIKE = 0.109) by 41.8% in this small sample. A larger-scale replication (K1139, planned) is required to determine whether the small-sample direction reverses at scale; until then, we regard the intraday frontier as theoretically plausible but empirically unconfirmed.
2. **Genuinely exogenous shocks:** Events that arrive between option-market pricing updates—geopolitical shocks, pandemic declarations, sudden policy announcements—may create temporary information asymmetries. However, these are by definition

unpredictable ex ante.

8.3 Economic Significance: Forecast Accuracy vs. Risk Management

A natural question is whether the forecast ranking differences documented in Section 7.6 translate into economically meaningful differences for risk management. We address this through a comprehensive VaR and Expected Shortfall (ES) backtesting exercise using five model families over 4,590 out-of-sample days (2008–2026), evaluated with the Kupiec (1995) unconditional coverage test, the Christoffersen (1998) conditional coverage (independence) test, and the Basel III traffic-light framework.

Table 11 presents the VaR/ES backtest results and the resulting economic ranking.

Table 11: VaR/ES Backtest: Model Risk Management Quality (2008–2026)

Model	VaR _{1%} Viol. Rate	Kupiec p	Christoff. p	Basel Zone	VaR _{5%} Viol. Rate	Kupiec p	Risk Score
AMEM(Gamma)	1.09%	0.549	0.576	Green	5.29%	0.365	1.94
GJR-GARCH(t)	1.35%	0.023	0.011	Green	4.58%	0.181	1.63
Hist. Sim(250d)	1.59%	<0.001	<0.001	Green	5.25%	0.440	1.34
EWMA(Gauss)	2.33%	<0.001	0.016	Yellow	5.90%	0.006	1.23
HAR-ABS(Gauss)	2.51%	<0.001	0.010	Yellow	6.36%	<0.001	1.19

Notes: Risk Score is a composite of Kupiec calibration, Christoffersen independence, Basel zone, and Acerbi-Szekely ES backtest, computed at both 1% and 5% levels (maximum score = 2.0). AMEM achieves near-perfect calibration at both levels (1.09% vs. 1% target; 5.29% vs. 5% target) and is the only model passing both Kupiec and Christoffersen at $\alpha = 0.01$. GJR-GARCH, despite being the best *forecaster* (Section 7.6), is only the second-best *risk manager*. All models use VIX as primary input; expanding window estimation with quarterly refits.

The key finding is a *decoupling* between forecast accuracy and risk management quality. GJR-GARCH is the best volatility forecaster (lowest QLIKE, sole MCS member), but

AMEM is the best risk manager (highest composite VaR/ES score of 1.94 vs. 1.63). This decoupling arises because risk management requires accurate *tail* calibration—getting the 1% and 5% quantiles right—which depends on the distributional assumption (AMEM’s Gamma distribution better captures the right-skewed nature of r_t^2) rather than on mean forecast accuracy alone.

For economic significance, we translate VaR/ES forecasts into portfolio allocations. An ES-weighted SPY/GLD strategy using AMEM achieves Sharpe = 0.861 versus the static 50/50 benchmark’s 0.814, an improvement of +0.047. While modest, this confirms that volatility models add value through risk control even when they cannot add value through return prediction—reinforcing the drawdown insurance framework.

Crucially, all five models in this exercise use VIX as their primary input signal, and none benefits from adding external signals. The debate over “which model is best” is entirely *within* the VIX information set, further strengthening VIX sufficiency: the relevant optimization is over model specification (GJR vs. AMEM) and application domain (forecasting vs. risk management), not over information sources.

8.4 The Drawdown Insurance Framework

If VT cannot generate alpha, what is its economic function? We propose that VT is best understood as drawdown insurance, analogous to buying a put option on the portfolio.

Proposition 1 (Drawdown Insurance Cost-Benefit). *For a CRRA investor with risk aversion γ , the certainty-equivalent return of a VT portfolio exceeds that of buy-and-hold when:*

$$\gamma \geq \gamma^* = \frac{\Delta\mu}{\frac{1}{2}(\sigma_{BH}^2 - \sigma_{VT}^2)}, \quad (14)$$

where $\Delta\mu = \mu_{BH} - \mu_{VT}$ is the return drag from VT, and $\sigma_{BH}^2 - \sigma_{VT}^2$ is the variance reduction.

Using our empirical estimates (Table 12), the 12/VIX strategy has a break-even risk aversion of $\gamma^* = 4.5$, and the EWMA VT strategy has $\gamma^* = 4.4$.

Table 12: VT as Drawdown Insurance: Cost-Benefit Analysis

VT Method	Drag	MDD Red.	Cost/1pp	γ^*	Welfare ($\gamma=5$)
12/VIX (SPY/GLD)	3.49%/yr	−8.2 pp	0.43%	4.5	+0.12%/yr
EWMA VT (SPY/GLD)	2.12%/yr	−6.1 pp	0.35%	4.4	+0.15%/yr

Notes: Return drag is the annualized difference in mean return between buy-and-hold and VT strategy. MDD reduction is the decrease in maximum drawdown. Cost per 1 pp MDD is the return drag divided by MDD improvement (in percentage points). Break-even γ is the minimum CRRA risk aversion at which the certainty-equivalent return of VT exceeds buy-and-hold. Welfare gain at $\gamma = 5$ is the annualized certainty-equivalent improvement. Cross-asset validation: 5 assets (SPY, GLD, QQQ, EEM, 0050.TW), 2007–2026.

The insurance framework has a clear practical implication: VT is rational for investors who are moderately to highly risk-averse ($\gamma \geq 4.5$, which encompasses most retail investors and many institutional mandates) but not for risk-neutral or mildly risk-averse investors who should prefer buy-and-hold.

8.5 The Simplicity Premium

A natural question is whether more complex strategies justify their complexity. Our meta-analysis of 14 live-tracked strategies reveals a *simplicity premium*: the Spearman correlation between the number of estimated parameters and the Sharpe ratio is $\rho = 0.077$ ($p = 0.794$), indicating zero relationship between complexity and performance. The best-performing strategies (50/50 SPY/GLD, 12/VIX, Piecewise VT) have zero or one estimated parameter.

This echoes DeMiguel et al.’s (2009) finding that $1/N$ portfolios outperform optimized alternatives. In the VT context, the simplicity premium arises because:

1. Simple rules have fewer parameters to estimate, reducing overfitting.
2. Smooth weight functions (like 12/VIX) produce low turnover, saving transaction

costs.

3. Zero-parameter rules are immune to estimation window and specification choices.

9 Conclusion

We set out to answer a straightforward question: can any observable signal beat VIX for equity volatility forecasting and volatility timing? After a systematic evaluation of eleven pre-specified signal families across 33 years of data, five market eras, and a unified pipeline with strict lag enforcement, transaction-cost accounting, Harvey (2016) approximate thresholds, formal Holm-Bonferroni multiple-testing correction, and proxy-robust evaluation per [Patton \(2011\)](#), the answer is no.

This null is not a failure but an informative finding. It reflects the efficiency of the options market, which aggregates heterogeneous information—cross-asset correlations, macroeconomic indicators, sentiment, term structure dynamics—into a single summary statistic. Each of the eleven signal families we tested represents one channel of information that option traders already incorporate into their pricing. VIX, as the price-weighted consensus, already contains them.

Moreover, the robustness of VIX sufficiency is underscored by three additional findings. First, volatility model rankings are criterion-dependent: GJR-GARCH dominates under proxy-robust QLIKE on r_t^2 , but AMEM dominates under $|r_t|$ targets. This instability in model rankings makes the stability of VIX sufficiency across all criteria all the more striking. Second, forecast accuracy and risk management quality are distinct dimensions—AMEM is the best risk manager (VaR/ES composite score 1.94) while GJR is the best forecaster—yet both operate entirely within the VIX information set, confirming that the relevant optimization is over model specification and application domain, not over information sources. Third, a seven-asset cross-asset PIT panorama (Table 9) delivers UNIVERSAL_NULL_7/7: of 28 specification cells spanning equities, commodities, fixed income, and cryptocurrency, only one isolated outlier reaches the Harvey threshold and it fails every robustness check, establishing native-IV sufficiency as the broadest

cross-asset finding at daily frequency.

The practical implications are clear:

1. **For researchers:** The daily-frequency, closing-price frontier is exhausted. Gains require genuinely exogenous information outside the option-market information set; the 5-minute realized-measure frontier is theoretically plausible but is not yet empirically confirmed (our K745 pilot with $N = 37$ found daily HAR-ABS *outperformed* 5-minute HAR-RV by 41.8%, a result that should be tested at scale before positing 5-minute data as a reliable improvement channel).
2. **For practitioners:** The simplest VIX-based rule (12/VIX) cannot be improved by adding signals. Complexity reduces returns through estimation error, higher turnover, and transaction costs. The optimal allocation for most investors is either buy-and-hold 50/50 SPY/GLD (if $\gamma < 4.5$) or 12/VIX VT (if $\gamma \geq 4.5$).
3. **For regulators and risk managers:** VIX sufficiency is time-invariant (CV = 0.37 across 33 years). Models based on VIX do not need frequent recalibration, and regulatory frameworks (e.g., Basel III/IV) that reference VIX for market risk are on solid empirical ground.

We conclude with a broader observation. In financial econometrics, null results are underreported relative to their information content. Our systematic horse race, with its pre-specified signal families and stringent statistical controls, provides stronger evidence than any individual positive finding could. The 36 null results accumulated across this research program are not 36 failures; they are 36 independent confirmations that the option market efficiently aggregates volatility-relevant information.

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